13. Diagonalization

Lecture due Feb 14, 2024 20:30 IST

Review: matrix with independent columns

1/1 point (graded)

Let $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n$ be a basis of $\mathbb{R}^n$ and let

$$\mathbf{S} = \begin{pmatrix} | & | & & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & & | \end{pmatrix} \tag{3.31}$$

where each vector $\mathbf{v}_i$ is placed in the i^{th} column of the matrix.

Which of the following is true?
(Choose all that apply.)

- [x] $\mathbf{S}$ is a square matrix.
- [x] $\mathbf{S}$ is invertible.
- [x] The null space is trivial: $\text{NS}\,(\mathbf{S}) = \mathbf{0}$.
- [x] The column space is all of $\mathbb{R}^n$: $\text{CS}\,(\mathbf{S}) = \mathbb{R}^n$.
- [ ] $\det\,(\mathbf{S}) = 0$
- [x] $\text{rank}\,(\mathbf{S}) = n$.

✔

Solution:

$\mathbf{S}$ is a square matrix because each $\mathbf{v}_i$ is a vector in $\mathbb{R}^n$ and there are n columns.
Since the set of all $\mathbf{v}_i$'s are linearly independent, $\mathbf{S}$ is invertible and has trivial null space: $\text{NS}\,(\mathbf{S}) = \mathbf{0}$. Since the set of $\mathbf{v}_i$'s spans $\mathbb{R}^n$, we have $\text{CS}\,(\mathbf{S}) = \mathbb{R}^n$ and $\text{rank}\,(\mathbf{S}) = \dim\text{CS}\,(\mathbf{S}) = n$.

Submit You have used 1 of 3 attempts

Answers are displayed within the problem

Diagonalization

Now, S inverse is coming from the right.

So can you keep those two straight?

A multiplies its eigenvectors, that's

how I keep them straight.

So A multiplies S. A multiplies S.

And then, this S inverse makes the whole thing diagonal.

And this is another way of saying the same thing,

putting the S's on the other side of the equation.

A is S Lambda S inverse.

So that's the new factorization.

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Suppose that $\mathbf{A}$ is a 2×2 matrix with a basis of eigenvectors $\mathbf{v}_1, \mathbf{v}_2$ corresponding to the distinct eigenvalues λ_1, λ_2 respectively (i.e. $\mathbf{A}$ is complete).

Warning : For what we are about to do, the eigenvectors must be listed in the same order as their eigenvalues.

Calculator

Use the eigenvalues to define a diagonal matrix $\mathbf{D}$:

$$\mathbf{D} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}.$$

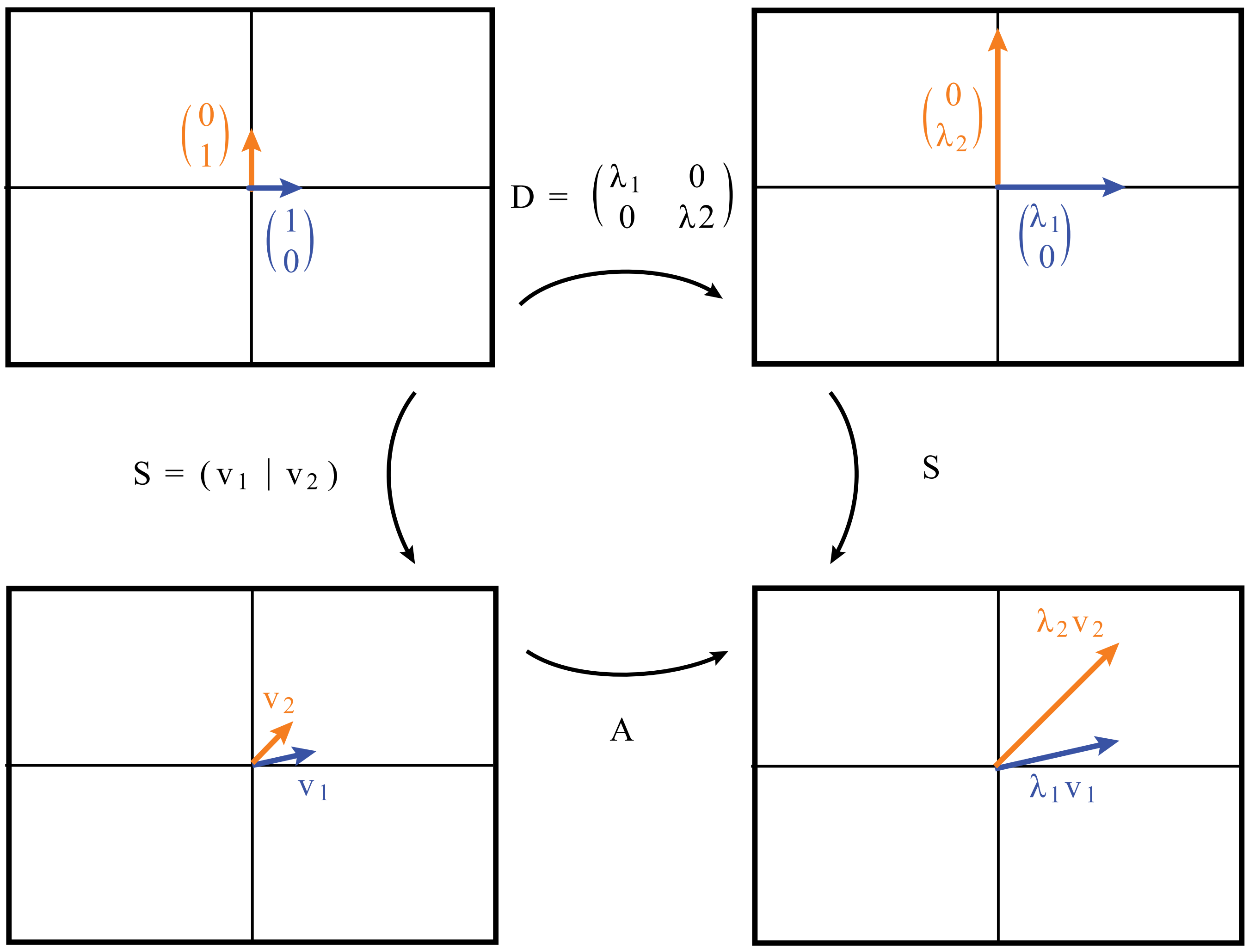
The matrix mapping $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ to $\mathbf{v}_1, \mathbf{v}_2$, respectively, is the matrix

$$\mathbf{S} = \begin{pmatrix} | & | \\ \mathbf{v}_1 & \mathbf{v}_2 \\ | & | \end{pmatrix}$$

whose columns are the eigenvectors $\mathbf{v}_1, \mathbf{v}_2$.

Question 13.1 What is the 2×2 matrix that maps $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ to $\lambda_1 \mathbf{v}_1, \lambda_2 \mathbf{v}_2$, respectively?

The picture below illustrates two ways of getting $\lambda_1 \mathbf{v}_1, \lambda_2 \mathbf{v}_2$ out of $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ respectively. Based on that image, construct two matrices that answer the question.



Answer 1

$\mathbf{AS}$, because applying $\mathbf{AS}$ means that $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are mapped by $\mathbf{S}$ to $\mathbf{v}_1, \mathbf{v}_2$, which are then mapped by $\mathbf{A}$ to $\lambda_1 \mathbf{v}_1, \lambda_2 \mathbf{v}_2$.

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Answer 2

$\mathbf{SD}$, because $\begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ are mapped by $\mathbf{D}$ to $\begin{pmatrix} \lambda_1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ \lambda_2 \end{pmatrix}$ which are then mapped by $\mathbf{S}$ to $\lambda_1 \mathbf{v}_1, \lambda_2 \mathbf{v}_2$.

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Conclusion: $\mathbf{AS} = \mathbf{SD}$.
(*Memory aid:* Just as in the eigenvalue-eigenvector equation $\mathbf{Av} = \lambda \mathbf{v}$ where $\mathbf{A}$ is applied to an eigenvector $\mathbf{v}$, here $\mathbf{A}$ should be applied to the matrix whose columns are the eigenvectors, hence $\mathbf{S}$.)

Multiply by $\mathbf{S}^{-1}$ on the right to get another way to write it:

$$\mathbf{A} = \mathbf{SDS}^{-1} \quad \text{where } \mathbf{S} = \begin{pmatrix} | & | \\ \mathbf{v}_1 & \mathbf{v}_2 \\ | & | \end{pmatrix} \tag{3.32}$$

Writing the matrix $\mathbf{A}$ like this is called **diagonalizing $\mathbf{A}$**.

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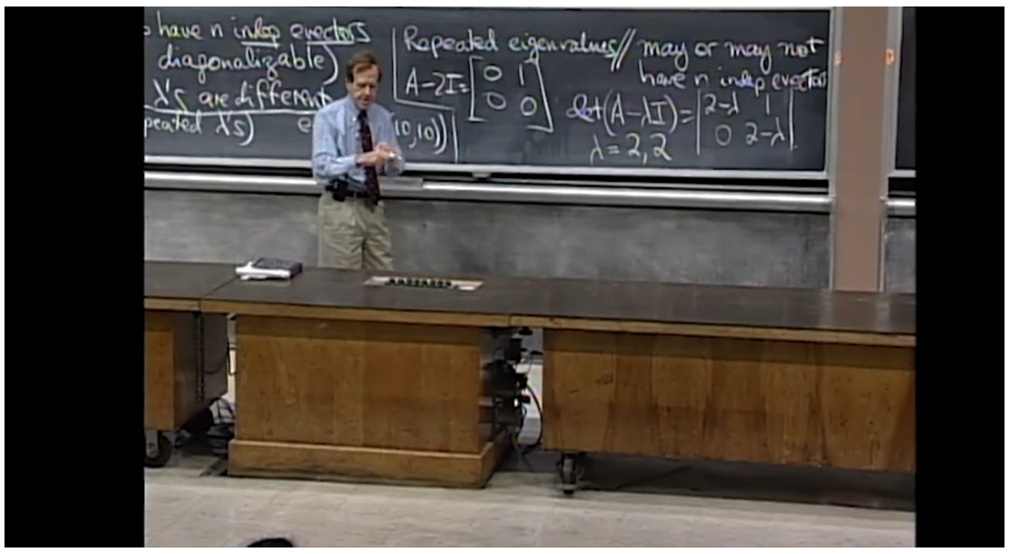
Remark:. We can think of the matrix $\mathbf{S} = (\mathbf{v}_1 \quad \mathbf{v}_2)$ as a "coordinate-change matrix" or "change-of-basis matrix" that

- relates the standard basis $\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}$ to the basis of eigenvectors of $\mathbf{A}$, and



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What are the eigenvectors?

They're the null space of A minus lambda I.

The null space is only one dimension.

This is a case where I don't have enough eigenvectors.

My algebraic multiplicity is 2.

When I count how often the eigenvalue is repeated,

that's the algebraic multiplicity.

That's the multiplicity-- how many times is

it the root of the polynomial?

My polynomial to 2 minus lambda squared.

It's a double root.

So my algebraic multiplicity is 2.

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